

# Sundry Asset Coverage and the Cross Section of Stock Returns

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## Abstract

This paper studies the asset pricing implications of Sundry Asset Coverage (SAC), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on SAC achieves an annualized gross (net) Sharpe ratio of 0.32 (0.28), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 22 (20) bps/month with a t-statistic of 3.23 (2.91), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Book to market, original (Statman 1980), Book leverage (annual), Operating leverage, Equity Duration, Enterprise Multiple, Earnings-to-Price Ratio) is 18 bps/month with a t-statistic of 2.63.

# 1 Introduction

Financial markets continuously process vast amounts of information, yet the speed and completeness of this incorporation remains a central question in asset pricing. While the efficient market hypothesis suggests that prices should rapidly reflect all available information, mounting evidence demonstrates that information diffuses gradually through markets, creating predictable patterns in the cross-section of stock returns. This slow diffusion manifests particularly in accounting-based signals, where complex financial statement information requires time and attention to fully incorporate into prices.

Despite extensive research on how investors process earnings, revenues, and major balance sheet items, we know surprisingly little about how markets incorporate information from sundry assets—the miscellaneous and often overlooked components of firm balance sheets. These sundry assets, while individually small, collectively represent material claims on firm value and contain information about managerial efficiency, operational flexibility, and future cash flows. The lack of attention to these assets creates a natural laboratory for examining how information complexity and investor inattention interact to generate return predictability in financial markets.

The slow diffusion of information provides a compelling framework for understanding why Sundry Asset Coverage (SAC) predicts future stock returns. Following the seminal work of [Hong et al. \(2000\)](#) on gradual information diffusion and [Hirshleifer and Teoh \(2003\)](#) on limited investor attention, we argue that information about sundry assets diffuses slowly through the market because these assets receive minimal analyst coverage and investor scrutiny. Unlike prominent balance sheet items that attract immediate attention, sundry assets require careful analysis to decode their implications for firm value, creating systematic underreaction to the information they contain.

The mechanism linking SAC to returns operates through investor inattention and

processing constraints. [DellaVigna and Pollet \(2009\)](#) demonstrate that investors systematically neglect information that appears less salient or requires additional cognitive effort to process. Sundry assets exemplify such overlooked information—they appear in footnotes and supplementary disclosures rather than headline numbers, require aggregation across multiple categories to assess their full impact, and lack standardized reporting formats that would facilitate easy comparison across firms. This complexity means that even sophisticated investors may initially overlook or underweight the information contained in sundry asset positions, leading to predictable drift as the market gradually incorporates this information.

The underreaction to SAC information should concentrate among firms where information frictions are most severe. [Zhang \(2006\)](#) shows that information uncertainty amplifies the market’s sluggish response to new information, while [Hou and Moskowitz \(2005\)](#) documents stronger return predictability among stocks with lower institutional ownership and analyst coverage. We therefore expect SAC to predict returns most strongly among smaller firms, those with limited analyst following, and stocks held primarily by retail investors who lack the resources to fully process complex accounting information. This cross-sectional variation in return predictability provides a crucial test of the information diffusion hypothesis and distinguishes it from risk-based explanations that would predict uniform effects across all stocks.

We find strong evidence that SAC predicts stock returns in the cross-section. A value-weighted long-short trading strategy based on SAC achieves an annualized gross Sharpe ratio of 0.32, with a monthly average excess return of 19 basis points ( $t$ -statistic = 2.49). The strategy’s performance remains robust after controlling for known risk factors, generating a monthly alpha of 22 basis points ( $t$ -statistic = 3.23) relative to the Fama-French six-factor model. These results establish SAC as an economically meaningful predictor of returns that cannot be explained by exposure to standard risk factors.

The economic magnitude and statistical significance of the SAC strategy persist across various portfolio construction methodologies and after accounting for transaction costs. Using alternative sorting procedures including name breaks and market capitalization breaks, the strategy maintains statistically significant alphas ranging from 22 to 24 basis points per month. After incorporating transaction costs using the composite bid-ask spreads from [Chen and Velikov \(2022\)](#), the net Sharpe ratio remains an economically meaningful 0.28, with a net six-factor alpha of 20 basis points per month (t-statistic = 2.91). The strategy’s net generalized alphas remain positive and significant across all standard factor models, demonstrating that SAC expands the investable opportunity set even after accounting for implementation costs.

Consistent with the information diffusion hypothesis, we document substantial variation in SAC’s predictive power across the size spectrum. Among the largest quintile of stocks—those above the 80th NYSE size percentile—the SAC strategy generates an average return of 15 basis points per month (t-statistic = 1.82) with a six-factor alpha of 20 basis points (t-statistic = 2.50). The strategy performs even more strongly among smaller stocks where information frictions are more severe, though the effect remains economically and statistically significant even among large-cap stocks where limits to arbitrage are minimal. This pattern supports our hypothesis that SAC captures gradual information incorporation rather than compensation for systematic risk.

Our study makes several contributions to the literature on information diffusion and cross-sectional return predictability. While [Sloan \(1996\)](#) demonstrates that investors fail to fully incorporate accruals information and [Richardson et al. \(2005\)](#) extends this finding to comprehensive balance sheet information in the *Journal of Accounting Research*, we provide the first systematic evidence that sundry assets—previously dismissed as immaterial—contain value-relevant information that predicts returns. Our findings complement [Hirshleifer et al. \(2004\)](#) in the *Journal*

of Accounting and Economics by showing that even within the balance sheet, investor attention varies dramatically across components, with sundry assets receiving particularly limited scrutiny despite their collective economic importance.

Methodologically, we advance the literature by demonstrating that SAC maintains its predictive power even after controlling for closely related anomalies including book-to-market, operating leverage, and earnings-to-price ratios. The SAC strategy generates a significant alpha of 18 basis points per month ( $t$ -statistic = 2.63) even after controlling for the Fama-French six factors plus six closely related strategies, establishing that SAC captures distinct information not subsumed by existing predictors. Following the comprehensive testing protocol of [Novy-Marx and Velikov \(2023\)](#), we show that SAC ranks in the top third of all documented anomalies for gross Sharpe ratio and, remarkably, in the top 11% for net performance after transaction costs, demonstrating unusual robustness to implementation frictions.

Our findings have broader implications for understanding how markets process complex accounting information and the limits of arbitrage in eliminating pricing inefficiencies. The persistence of the SAC anomaly even among large-cap stocks suggests that information complexity, rather than just transaction costs or limits to arbitrage, plays a crucial role in generating cross-sectional return predictability. These results support theoretical models of limited attention such as [Peng and Xiong \(2006\)](#) in the *Journal of Finance* and provide empirical evidence for the gradual information diffusion framework of [Hong and Stein \(1999\)](#) in the *Review of Financial Studies*, while highlighting the economic importance of seemingly minor accounting items in the price discovery process.

## 2 Data

We obtain financial statement data from the COMPUSTAT North America database, which provides comprehensive accounting information for publicly traded U.S. companies. The database offers standardized financial statement items that enable consistent measurement across firms and over time. Our sample includes all non-financial firms with available data spanning several decades, providing broad coverage of the U.S. equity market. Sundry Asset Coverage signal utilizes two key accounting variables. Current assets other sundry (ACOX) represents miscellaneous current assets not classified in standard current asset categories such as cash, receivables, or inventory. These sundry assets typically include prepaid expenses, deferred charges, and other short-term assets that do not fit into conventional classifications. Equity liquidating value (CEQL) measures the liquidating value of common equity, representing the amount that would theoretically be available to common shareholders if the firm's assets were liquidated at their book values and all liabilities were satisfied. We construct the Sundry Asset Coverage signal by dividing current assets other sundry (ACOX) by equity liquidating value (CEQL). This ratio captures the proportion of miscellaneous current assets relative to the liquidation value available to equity holders, providing insight into the composition and quality of a firm's asset base. We calculate this measure using annual observations from fiscal year-end financial statements. To ensure data quality, we require non-missing values for both ACOX and CEQL, and exclude observations where CEQL equals zero to avoid undefined ratios. We winsorize the resulting signal at the 1st and 99th percentiles to mitigate the influence of extreme outliers.

### 3 Signal diagnostics

Figure 1 plots descriptive statistics for the SAC signal. Panel A plots the time-series of the mean, median, and interquartile range for SAC. On average, the cross-sectional mean (median) SAC is 0.07 (0.02) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input SAC data. The signal's interquartile range spans -0.00 to 0.10. Panel B of Figure 1 plots the time-series of the coverage of the SAC signal for the CRSP universe. On average, the SAC signal is available for 6.30% of CRSP names, which on average make up 7.31% of total market capitalization.

### 4 Does SAC predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on SAC using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high SAC portfolio and sells the low SAC portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short SAC strategy earns an average return of 0.19% per month with a t-statistic of 2.49. The annualized Sharpe ratio of the strategy is 0.32. The alphas range from 0.20% to 0.31% per month and have t-statistics exceeding 2.51 everywhere. The lowest alpha is with respect to the CAPM factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is -0.38,

with a t-statistic of -12.00 on the HML factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 559 stocks and an average market capitalization of at least \$1,537 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals 17 bps/month with a t-statistics of 3.11. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-two exceed two, and for thirteen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns



reported in the first column range between -1-19bps/month. The lowest return, (-1 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of -0.11. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the SAC trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty cases, and significantly expands the achievable frontier in nineteen cases.

Table 3 provides direct tests for the role size plays in the SAC strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and SAC, as well as average returns and alphas for long/short trading SAC strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80<sup>th</sup> NYSE percentile), the SAC strategy achieves an average return of 15 bps/month with a t-statistic of 1.82. Among these large cap stocks, the alphas for the SAC strategy relative to the five most common factor models range from 17 to 29 bps/month with t-statistics between 2.03 and 3.86.

## 5 How does SAC perform relative to the zoo?

Figure 2 puts the performance of SAC in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.<sup>1</sup> The vertical red line shows where the Sharpe ratio for the SAC strategy falls in the distribution. The SAC strategy’s gross (net) Sharpe ratio of 0.32 (0.28) is greater than 67% (89%) of anomaly Sharpe ratios,

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<sup>1</sup>The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the SAC strategy (red line).<sup>2</sup> Ignoring trading costs, a \$1 invested in the SAC strategy would have yielded \$2.59 which ranks the SAC strategy in the top 11% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the SAC strategy would have yielded \$1.99 which ranks the SAC strategy in the top 7% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the SAC relative to those. Panel A shows that the SAC strategy gross alphas fall between the 42 and 72 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The SAC strategy has a positive net generalized alpha for five out of the five factor models. In these cases SAC ranks between the 64 and 89 percentiles in terms of how much it could have expanded the achievable investment frontier.

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<sup>2</sup>The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

## 6 Does SAC add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of SAC with 210 filtered anomaly signals.<sup>3</sup> Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price SAC or at least to weaken the power SAC has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of SAC conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on  $\beta_{SAC}$  from Fama-MacBeth regressions of the form  $r_{i,t} = \alpha + \beta_{SAC}SAC_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$ , where  $X$  stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on  $\alpha$  from spanning tests of the form:  $r_{SAC,t} = \alpha + \beta r_{X,t} + \epsilon_t$ , where  $r_{X,t}$  stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on SAC. Stocks are finally grouped into five SAC portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted

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<sup>3</sup>When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

SAC trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on SAC and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the  $R^2$  from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the SAC signal in these Fama-MacBeth regressions exceed 1.91, with the minimum t-statistic occurring when controlling for Operating leverage. Controlling for all six closely related anomalies, the t-statistic on SAC is 2.62.

Similarly, Table 5 reports results from spanning tests that regress returns to the SAC strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the SAC strategy earns alphas that range from 20-25bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.88, which is achieved when controlling for Operating leverage. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the SAC trading strategy achieves an alpha of 18bps/month with a t-statistic of 2.63.

## 7 Does SAC add relative to the whole zoo?

Finally, we can ask how much adding SAC to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the SAC signal.<sup>4</sup> We consider one different methods for combining signals.

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<sup>4</sup>We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which SAC is available.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 154-anomaly combination strategy grows to \$2560.30, while \$1 investment in the combination strategy that includes SAC grows to \$2537.65.

## 8 Conclusion

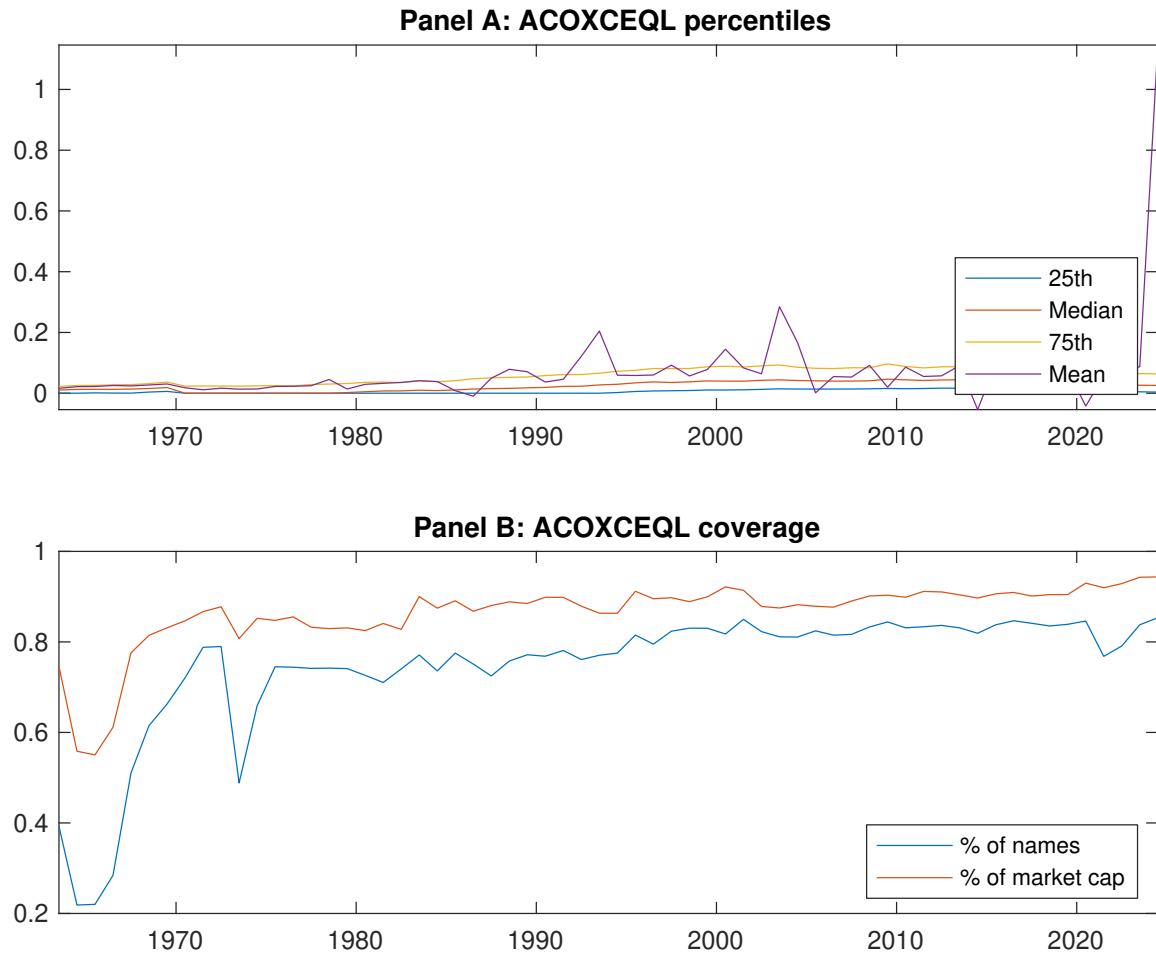
This study provides compelling evidence that Sundry Asset Coverage (SAC) represents a robust and economically significant predictor of cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that SAC-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for well-established factors and closely related signals from the factor zoo.

The key findings of our research underscore the economic importance of SAC as an asset pricing signal. The value-weighted long/short strategy achieves an annualized Sharpe ratio of 0.32 gross (0.28 net of transaction costs), indicating strong performance that remains viable after accounting for implementation frictions. More notably, the strategy generates statistically significant abnormal returns of 22 basis points per month (t-statistic = 3.23) relative to the Fama-French five-factor model augmented with momentum, with net returns remaining economically meaningful at 20 basis points monthly. The robustness of these results is further validated by the persistence of significant alpha (18 bps/month, t-statistic = 2.63) even after controlling for six closely related strategies, including traditional value measures, leverage metrics, and earnings-based ratios.

From a practical perspective, these findings suggest that SAC captures unique information about firm fundamentals not fully reflected in existing factor models or related accounting-based signals. The strategy’s ability to generate positive net returns after transaction costs indicates potential implementability for institutional investors, while its orthogonality to established factors suggests diversification benefits when incorporated into multi-factor portfolios.

Despite these encouraging results, several limitations warrant consideration. First, while we employ the Novy-Marx and Velikov (2023) protocol to ensure robustness, the analysis remains subject to potential data-mining concerns inherent in factor discovery research. Second, the study focuses primarily on U.S. equity markets, and the international applicability of SAC remains unexplored. Third, while transaction cost estimates follow standard conventions, actual implementation costs may vary significantly across different market conditions and investor types.

Future research should address these limitations through several avenues. International out-of-sample tests would provide valuable evidence on the global robustness of the SAC signal. Time-series analysis examining the strategy’s performance across different market regimes and economic cycles could illuminate the underlying economic mechanisms driving the premium. Additionally, investigating the interaction between SAC and corporate events such as mergers, restructurings, or accounting changes may provide deeper insights into the information content captured by this measure. Finally, exploring the optimal combination of SAC with other factors in portfolio construction represents a promising direction for enhancing practical implementation strategies.



**Figure 1:** Times series of SAC percentiles and coverage.  
This figure plots descriptive statistics for SAC. Panel A shows cross-sectional percentiles of SAC over the sample. Panel B plots the monthly coverage of SAC relative to the universe of CRSP stocks with available market capitalizations.

**Table 1:** Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on SAC. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

| Panel A: Excess returns and alphas on SAC-sorted portfolios                       |                  |                  |                  |                  |                  |                   |
|-----------------------------------------------------------------------------------|------------------|------------------|------------------|------------------|------------------|-------------------|
|                                                                                   | (L)              | (2)              | (3)              | (4)              | (H)              | (H-L)             |
| $r^e$                                                                             | 0.56<br>[3.14]   | 0.51<br>[2.87]   | 0.63<br>[3.72]   | 0.62<br>[3.69]   | 0.75<br>[4.29]   | 0.19<br>[2.49]    |
| $\alpha_{CAPM}$                                                                   | -0.05<br>[-0.90] | -0.10<br>[-1.96] | 0.05<br>[1.06]   | 0.04<br>[0.97]   | 0.15<br>[3.23]   | 0.20<br>[2.51]    |
| $\alpha_{FF3}$                                                                    | -0.13<br>[-2.86] | -0.07<br>[-1.47] | 0.07<br>[1.59]   | 0.09<br>[2.27]   | 0.19<br>[4.15]   | 0.31<br>[4.64]    |
| $\alpha_{FF4}$                                                                    | -0.11<br>[-2.50] | -0.04<br>[-0.76] | 0.04<br>[0.93]   | 0.08<br>[1.92]   | 0.17<br>[3.71]   | 0.28<br>[4.12]    |
| $\alpha_{FF5}$                                                                    | -0.15<br>[-3.32] | -0.02<br>[-0.37] | 0.05<br>[1.08]   | 0.08<br>[1.87]   | 0.09<br>[2.14]   | 0.24<br>[3.57]    |
| $\alpha_{FF6}$                                                                    | -0.14<br>[-2.98] | 0.01<br>[0.15]   | 0.03<br>[0.58]   | 0.07<br>[1.59]   | 0.08<br>[1.96]   | 0.22<br>[3.23]    |
| Panel B: Fama and French (2018) 6-factor model loadings for SAC-sorted portfolios |                  |                  |                  |                  |                  |                   |
| $\beta_{MKT}$                                                                     | 1.08<br>[96.74]  | 1.01<br>[83.14]  | 0.98<br>[92.87]  | 0.98<br>[96.65]  | 1.02<br>[98.95]  | -0.06<br>[-3.45]  |
| $\beta_{SMB}$                                                                     | -0.07<br>[-4.44] | -0.03<br>[-1.76] | 0.05<br>[3.37]   | -0.00<br>[-0.16] | 0.09<br>[6.18]   | 0.16<br>[6.85]    |
| $\beta_{HML}$                                                                     | 0.22<br>[10.17]  | -0.06<br>[-2.39] | -0.07<br>[-3.56] | -0.12<br>[-6.07] | -0.16<br>[-8.24] | -0.38<br>[-12.00] |
| $\beta_{RMW}$                                                                     | 0.05<br>[2.23]   | -0.12<br>[-5.16] | 0.04<br>[1.88]   | 0.06<br>[2.90]   | 0.22<br>[11.10]  | 0.18<br>[5.42]    |
| $\beta_{CMA}$                                                                     | 0.04<br>[1.42]   | -0.05<br>[-1.32] | 0.02<br>[0.71]   | -0.04<br>[-1.25] | 0.10<br>[3.34]   | 0.05<br>[1.13]    |
| $\beta_{UMD}$                                                                     | -0.02<br>[-1.77] | -0.04<br>[-3.11] | 0.03<br>[2.97]   | 0.02<br>[1.52]   | 0.01<br>[0.91]   | 0.03<br>[1.76]    |
| Panel C: Average number of firms ( $n$ ) and market capitalization ( $me$ )       |                  |                  |                  |                  |                  |                   |
| $n$                                                                               | 697              | 813              | 704              | 601              | 559              |                   |
| $me$ (\$10 <sup>6</sup> )                                                         | 1537             | 1623             | 1908             | 2200             | 2207             |                   |



**Table 2:** Robustness to sorting methodology & trading costs

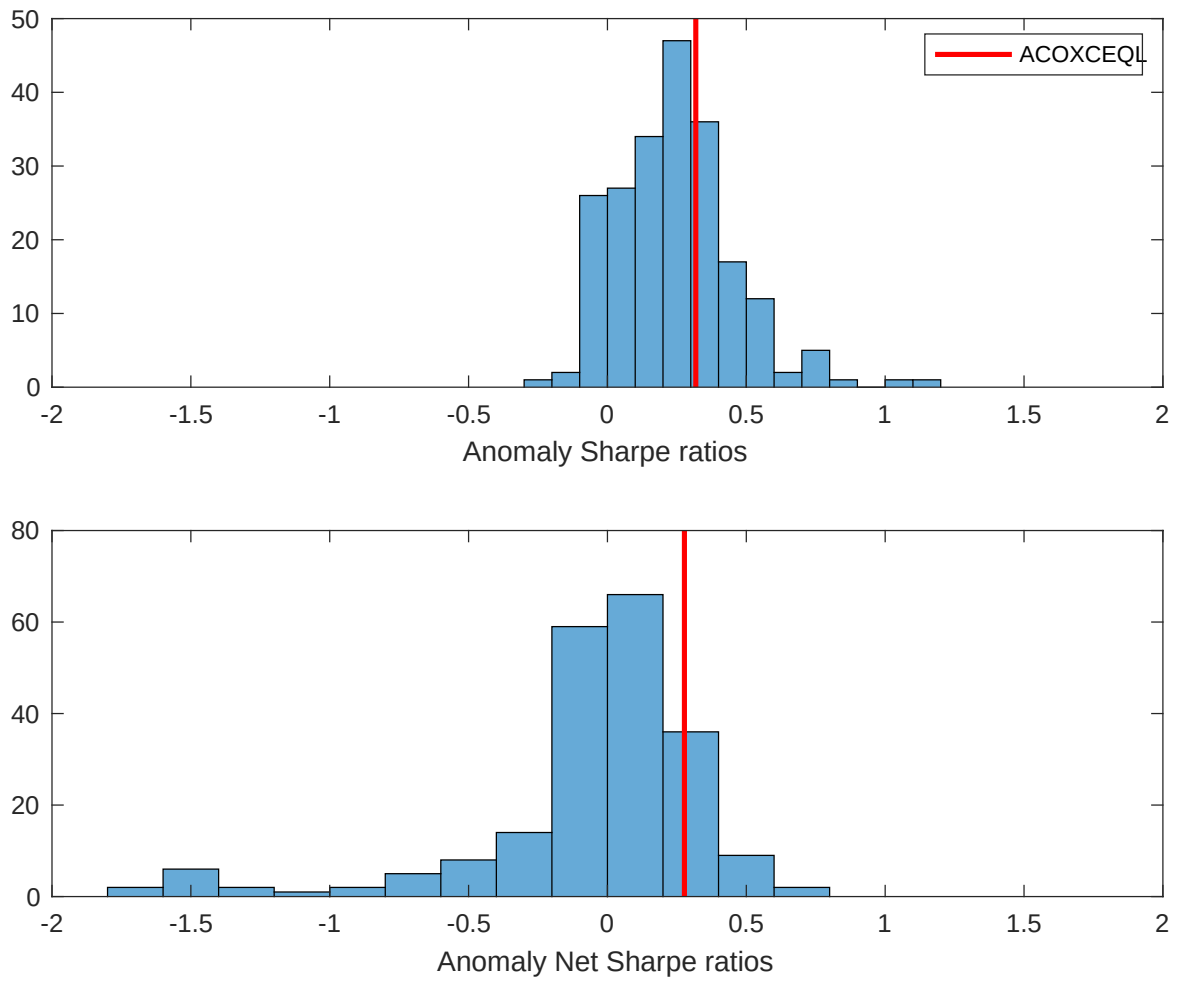
This table evaluates the robustness of the choices made in the SAC strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

| Panel A: Gross Returns and Alphas                                        |        |         |                    |                          |                         |                         |                         |                         |
|--------------------------------------------------------------------------|--------|---------|--------------------|--------------------------|-------------------------|-------------------------|-------------------------|-------------------------|
| Portfolios                                                               | Breaks | Weights | $r^e$              | $\alpha_{\text{CAPM}}$   | $\alpha_{\text{FF3}}$   | $\alpha_{\text{FF4}}$   | $\alpha_{\text{FF5}}$   | $\alpha_{\text{FF6}}$   |
| Quintile                                                                 | NYSE   | VW      | 0.19<br>[2.49]     | 0.20<br>[2.51]           | 0.31<br>[4.64]          | 0.28<br>[4.12]          | 0.24<br>[3.57]          | 0.22<br>[3.23]          |
| Quintile                                                                 | NYSE   | EW      | 0.17<br>[3.11]     | 0.14<br>[2.50]           | 0.12<br>[2.28]          | 0.12<br>[2.21]          | 0.05<br>[1.03]          | 0.06<br>[1.08]          |
| Quintile                                                                 | Name   | VW      | 0.22<br>[2.75]     | 0.22<br>[2.73]           | 0.34<br>[4.89]          | 0.30<br>[4.29]          | 0.26<br>[3.79]          | 0.24<br>[3.39]          |
| Quintile                                                                 | Cap    | VW      | 0.19<br>[2.48]     | 0.20<br>[2.60]           | 0.31<br>[4.56]          | 0.28<br>[4.08]          | 0.24<br>[3.57]          | 0.22<br>[3.26]          |
| Decile                                                                   | NYSE   | VW      | 0.18<br>[2.00]     | 0.18<br>[1.98]           | 0.27<br>[3.19]          | 0.25<br>[2.89]          | 0.25<br>[2.83]          | 0.23<br>[2.59]          |
| Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas |        |         |                    |                          |                         |                         |                         |                         |
| Portfolios                                                               | Breaks | Weights | $r_{\text{net}}^e$ | $\alpha_{\text{CAPM}}^*$ | $\alpha_{\text{FF3}}^*$ | $\alpha_{\text{FF4}}^*$ | $\alpha_{\text{FF5}}^*$ | $\alpha_{\text{FF6}}^*$ |
| Quintile                                                                 | NYSE   | VW      | 0.17<br>[2.17]     | 0.17<br>[2.21]           | 0.28<br>[4.08]          | 0.26<br>[3.81]          | 0.21<br>[3.09]          | 0.20<br>[2.91]          |
| Quintile                                                                 | NYSE   | EW      | -0.01<br>[-0.11]   |                          |                         |                         |                         |                         |
| Quintile                                                                 | Name   | VW      | 0.19<br>[2.44]     | 0.20<br>[2.45]           | 0.30<br>[4.35]          | 0.28<br>[4.04]          | 0.23<br>[3.34]          | 0.21<br>[3.12]          |
| Quintile                                                                 | Cap    | VW      | 0.16<br>[2.14]     | 0.17<br>[2.17]           | 0.26<br>[3.90]          | 0.25<br>[3.64]          | 0.20<br>[2.97]          | 0.19<br>[2.81]          |
| Decile                                                                   | NYSE   | VW      | 0.15<br>[1.62]     | 0.15<br>[1.58]           | 0.23<br>[2.65]          | 0.22<br>[2.49]          | 0.19<br>[2.20]          | 0.18<br>[2.09]          |

**Table 3:** Conditional sort on size and SAC

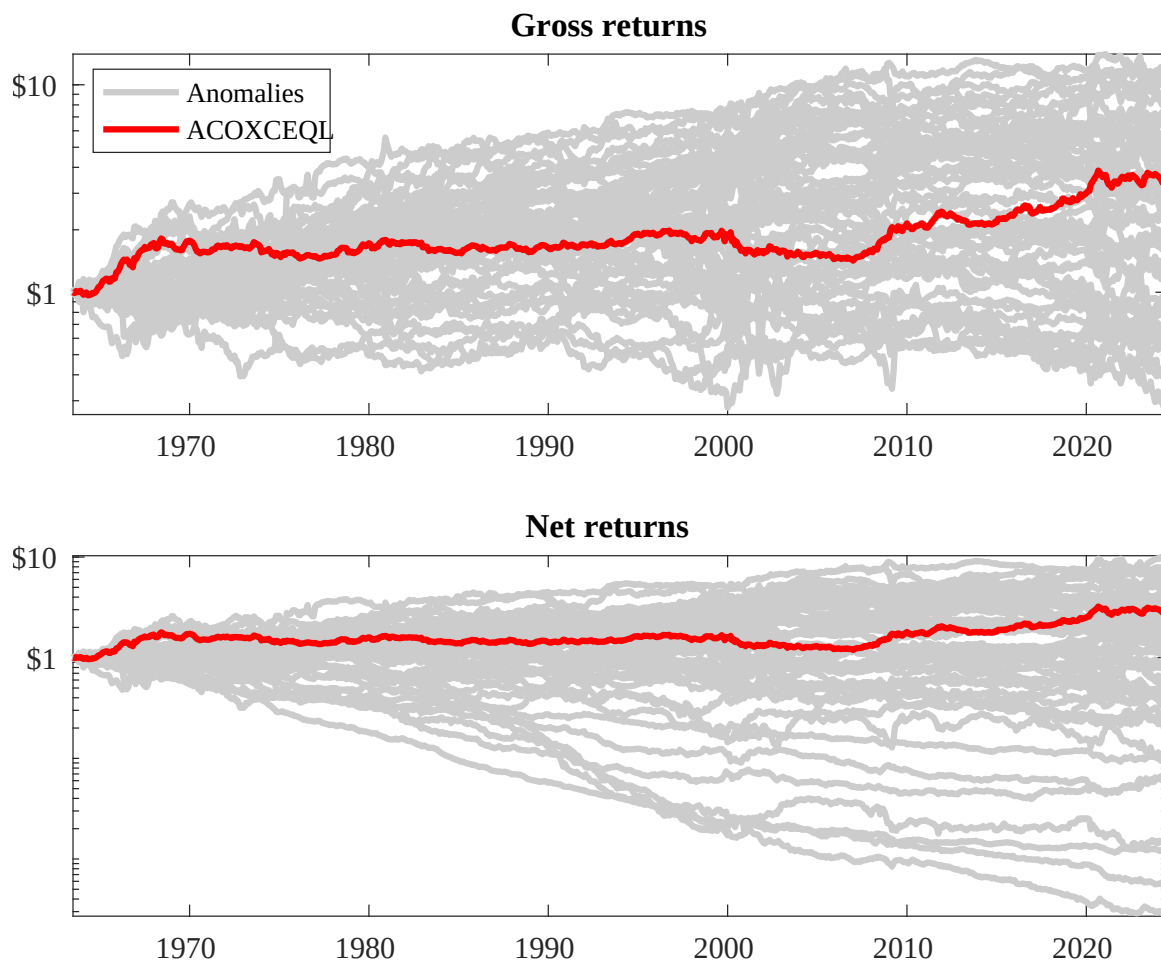
This table presents results for conditional double sorts on size and SAC. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on SAC. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high SAC and short stocks with low SAC. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 1963:06 to 2024:06.

| Panel A: portfolio average returns and time-series regression results |               |                |                |                |                |                |                                                    |                 |                |                |                |                |
|-----------------------------------------------------------------------|---------------|----------------|----------------|----------------|----------------|----------------|----------------------------------------------------|-----------------|----------------|----------------|----------------|----------------|
| Size quintiles                                                        | SAC Quintiles |                |                |                |                | SAC Strategies |                                                    |                 |                |                |                |                |
|                                                                       |               | (L)            | (2)            | (3)            | (4)            | (H)            | $r^e$                                              | $\alpha_{CAPM}$ | $\alpha_{FF3}$ | $\alpha_{FF4}$ | $\alpha_{FF5}$ | $\alpha_{FF6}$ |
|                                                                       | (1)           | 0.61<br>[2.42] | 0.54<br>[2.11] | 0.80<br>[3.14] | 0.88<br>[3.56] | 0.84<br>[3.03] | 0.23<br>[2.10]                                     | 0.19<br>[1.70]  | 0.14<br>[1.24] | 0.13<br>[1.15] | 0.07<br>[0.63] | 0.07<br>[0.62] |
|                                                                       | (2)           | 0.76<br>[3.26] | 0.65<br>[2.74] | 0.76<br>[3.20] | 0.81<br>[3.45] | 0.93<br>[3.82] | 0.17<br>[2.08]                                     | 0.14<br>[1.73]  | 0.13<br>[1.58] | 0.16<br>[1.92] | 0.09<br>[1.02] | 0.12<br>[1.36] |
|                                                                       | (3)           | 0.72<br>[3.42] | 0.65<br>[3.07] | 0.77<br>[3.54] | 0.82<br>[3.71] | 0.93<br>[4.11] | 0.21<br>[2.52]                                     | 0.16<br>[1.94]  | 0.16<br>[1.93] | 0.15<br>[1.81] | 0.13<br>[1.59] | 0.13<br>[1.53] |
|                                                                       | (4)           | 0.65<br>[3.19] | 0.57<br>[2.93] | 0.73<br>[3.61] | 0.90<br>[4.47] | 0.77<br>[3.74] | 0.12<br>[1.50]                                     | 0.11<br>[1.35]  | 0.17<br>[2.06] | 0.15<br>[1.80] | 0.11<br>[1.29] | 0.10<br>[1.16] |
|                                                                       | (5)           | 0.56<br>[3.17] | 0.55<br>[3.19] | 0.55<br>[3.42] | 0.54<br>[3.24] | 0.71<br>[4.23] | 0.15<br>[1.82]                                     | 0.17<br>[2.03]  | 0.29<br>[3.86] | 0.26<br>[3.37] | 0.22<br>[2.81] | 0.20<br>[2.50] |
| Panel B: Portfolio average number of firms and market capitalization  |               |                |                |                |                |                |                                                    |                 |                |                |                |                |
| Size quintiles                                                        | SAC Quintiles |                |                |                |                | SAC Quintiles  |                                                    |                 |                |                |                |                |
|                                                                       |               | Average $n$    |                |                |                |                | Average market capitalization (\$10 <sup>6</sup> ) |                 |                |                |                |                |
|                                                                       |               | (L)            | (2)            | (3)            | (4)            | (H)            | (L)                                                | (2)             | (3)            | (4)            | (H)            |                |
|                                                                       | (1)           | 378            | 386            | 388            | 387            | 380            | 28                                                 | 31              | 33             | 32             | 29             |                |
|                                                                       | (2)           | 104            | 104            | 104            | 104            | 103            | 53                                                 | 54              | 54             | 54             | 54             |                |
|                                                                       | (3)           | 72             | 72             | 73             | 73             | 72             | 92                                                 | 90              | 92             | 92             | 94             |                |
|                                                                       | (4)           | 60             | 60             | 60             | 60             | 60             | 199                                                | 192             | 196            | 197            | 204            |                |
| (5)                                                                   | 55            | 55             | 55             | 55             | 55             | 1347           | 1402                                               | 1470            | 1784           | 1604           |                |                |



**Figure 2:** Distribution of Sharpe ratios.

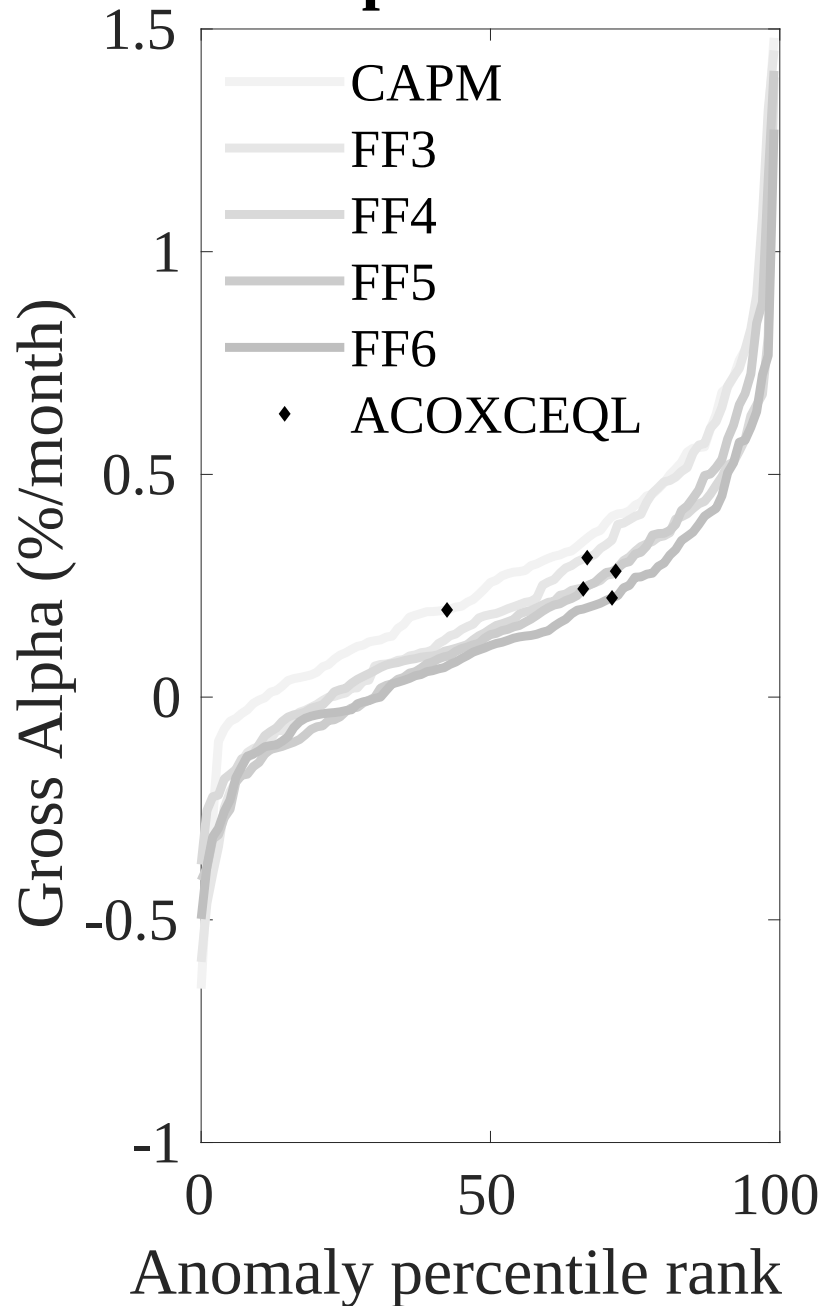
This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the SAC with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.



**Figure 3:** Dollar invested.

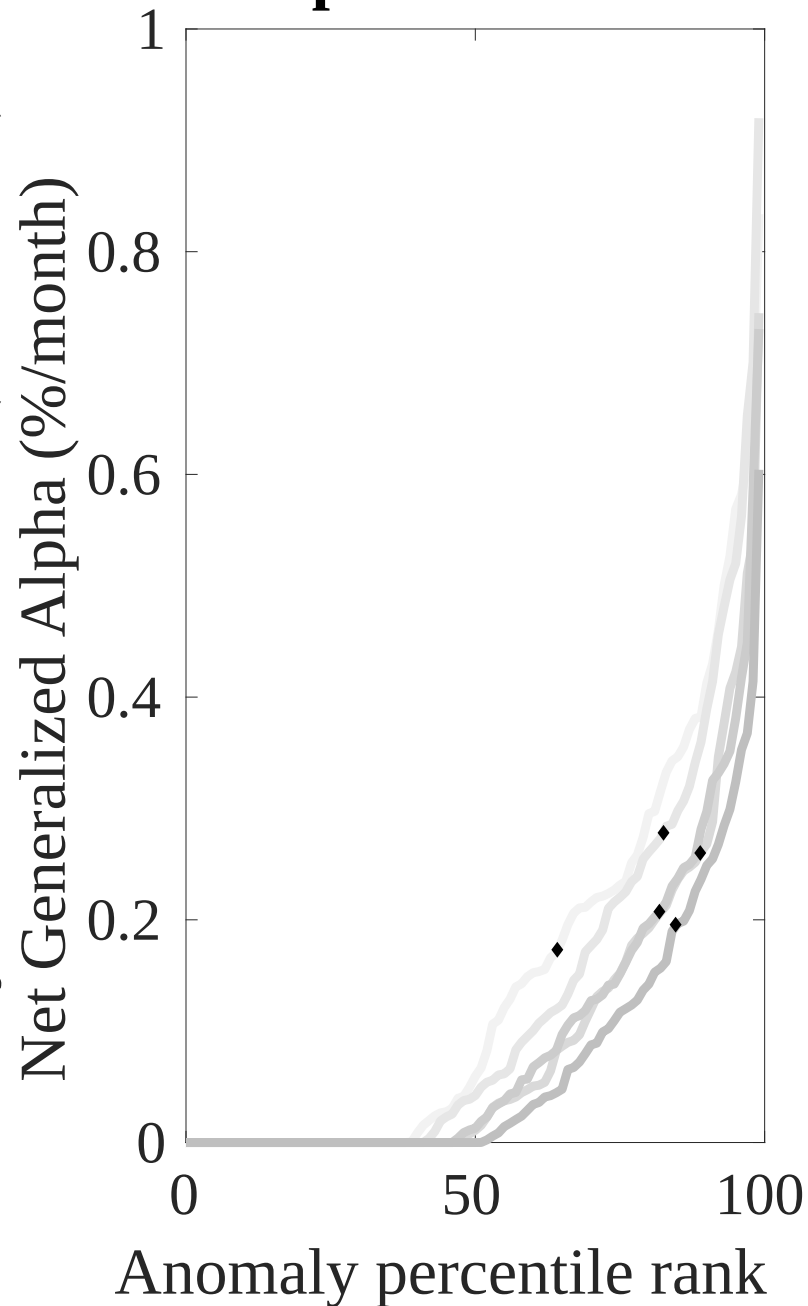
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the SAC trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

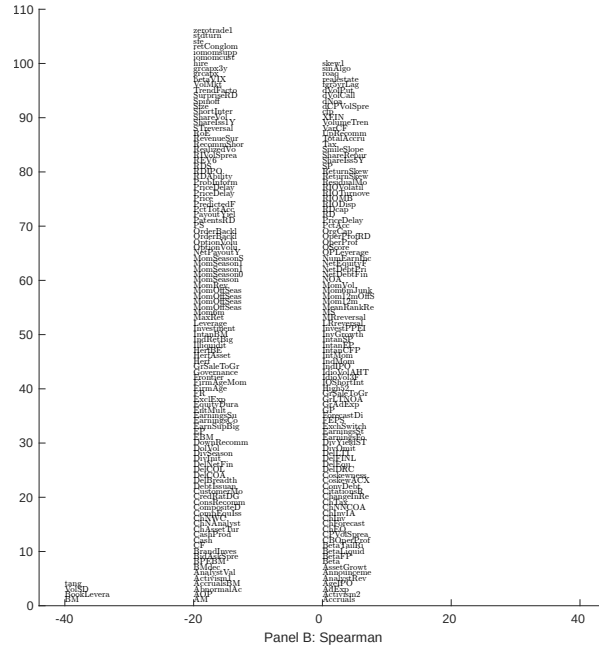
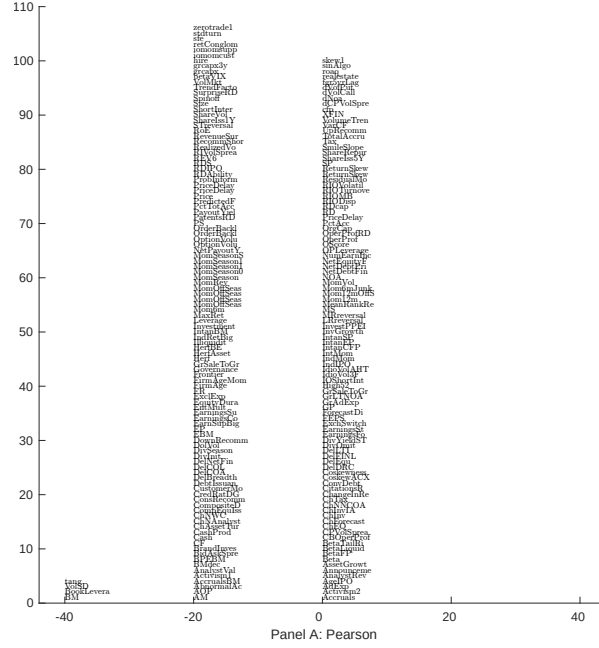
## Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

## Net Alpha Percentiles

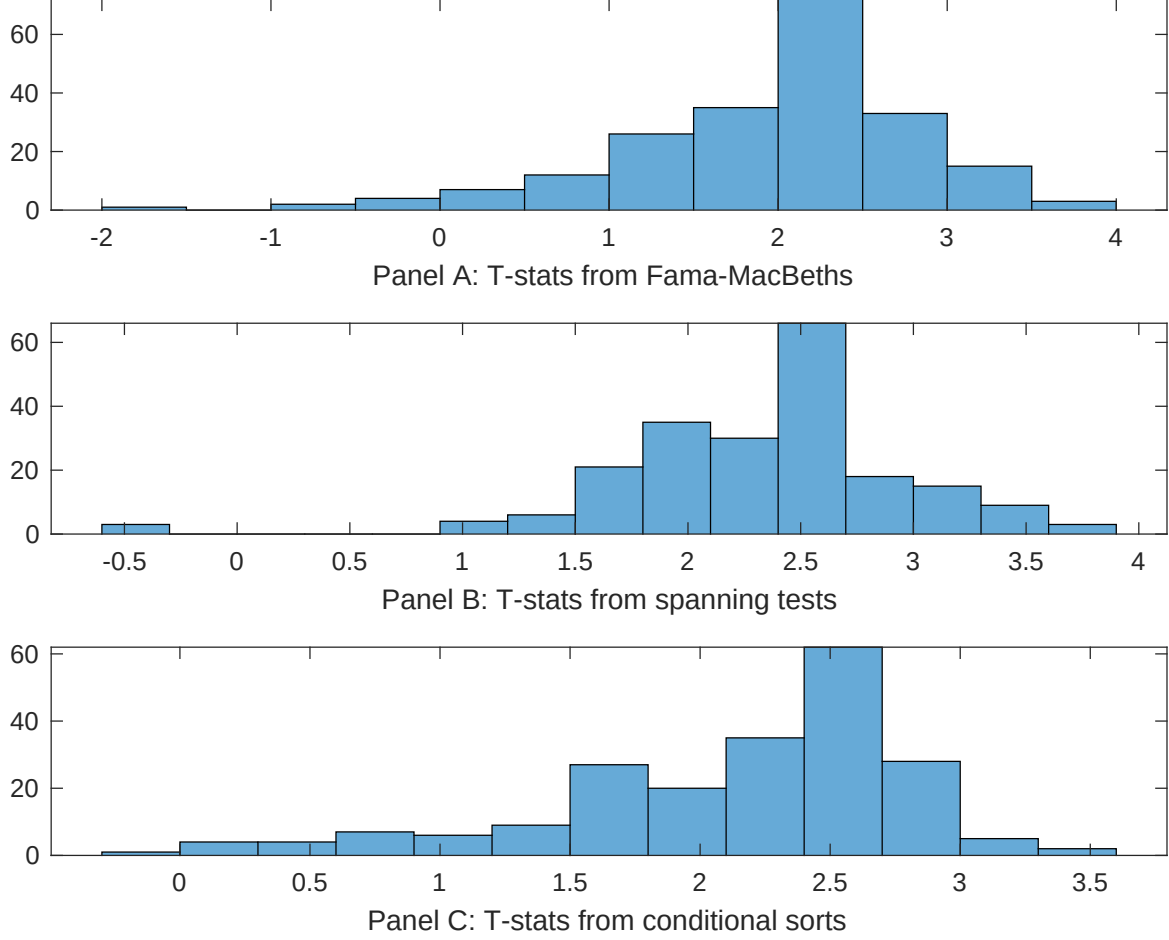




**Figure 5:** Distribution of correlations.

This figure plots a name histogram of correlations of 210 filtered anomaly signals with SAC. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.



**Figure 7:** Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of SAC conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on  $\beta_{SAC}$  from Fama-MacBeth regressions of the form  $r_{i,t} = \alpha + \beta_{SAC}SAC_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$ , where  $X$  stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on  $\alpha$  from spanning tests of the form:  $r_{SAC,t} = \alpha + \beta r_{X,t} + \epsilon_t$ , where  $r_{X,t}$  stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on SAC. Stocks are finally grouped into five SAC portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted SAC trading strategies conditioned on each of the 210 filtered anomalies.



**Table 4:** Fama-MacBeths controlling for most closely related anomalies

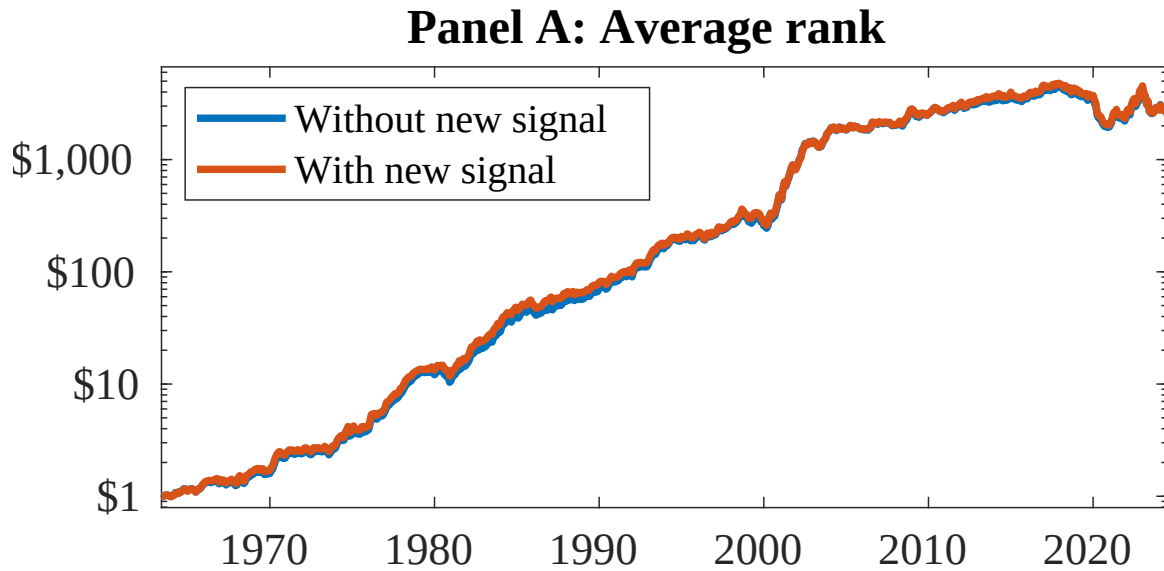
This table presents Fama-MacBeth results of returns on SAC. and the six most closely related anomalies. The regressions take the following form:  $r_{i,t} = \alpha + \beta_{SAC}SAC_{i,t} + \sum_{k=1}^s \beta_{X_k}X_{i,t}^k + \epsilon_{i,t}$ . The six most closely related anomalies,  $X$ , are Book to market, original (Stattman 1980), Book leverage (annual), Operating leverage, Equity Duration, Enterprise Multiple, Earnings-to-Price Ratio. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the  $R^2$  from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

|                 |                |                |                |                |                |                |                  |
|-----------------|----------------|----------------|----------------|----------------|----------------|----------------|------------------|
| Intercept       | 0.14<br>[6.32] | 0.12<br>[5.20] | 0.10<br>[4.47] | 0.20<br>[9.06] | 0.13<br>[6.31] | 0.11<br>[4.94] | 0.21<br>[8.42]   |
| SAC             | 0.11<br>[3.22] | 0.92<br>[3.23] | 0.60<br>[1.91] | 0.10<br>[3.29] | 0.78<br>[2.45] | 0.79<br>[2.60] | 0.78<br>[2.62]   |
| Anomaly 1       | 0.38<br>[7.71] |                |                |                |                |                | 0.90<br>[2.01]   |
| Anomaly 2       |                | 0.16<br>[1.55] |                |                |                |                | -0.17<br>[-1.19] |
| Anomaly 3       |                |                | 0.11<br>[3.12] |                |                |                | 0.87<br>[2.66]   |
| Anomaly 4       |                |                |                | 0.52<br>[5.34] |                |                | 0.58<br>[5.44]   |
| Anomaly 5       |                |                |                |                | 0.81<br>[3.47] |                | 1.00<br>[3.12]   |
| Anomaly 6       |                |                |                |                |                | 0.17<br>[2.57] | -0.45<br>[-0.65] |
| # months        | 727            | 732            | 732            | 720            | 727            | 727            | 715              |
| $\bar{R}^2(\%)$ | 1              | 0              | 0              | 1              | 0              | 1              | 0                |

**Table 5:** Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the SAC trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form:  $r_t^{SAC} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$ , where  $X_k$  indicates each of the six most-closely related anomalies and  $f_j$  indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies,  $X$ , are Book to market, original (Stattman 1980), Book leverage (annual), Operating leverage, Equity Duration, Enterprise Multiple, Earnings-to-Price Ratio. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the  $R^2$  from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

|                 |                   |                   |                   |                   |                   |                   |                   |
|-----------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|
| Intercept       | 0.20<br>[2.88]    | 0.20<br>[2.94]    | 0.20<br>[2.92]    | 0.22<br>[3.16]    | 0.25<br>[3.60]    | 0.22<br>[3.17]    | 0.18<br>[2.63]    |
| Anomaly 1       | -21.13<br>[-5.34] |                   |                   |                   |                   |                   | -20.81<br>[-4.60] |
| Anomaly 2       |                   | 6.86<br>[2.67]    |                   |                   |                   |                   | 4.38<br>[1.60]    |
| Anomaly 3       |                   |                   | 18.48<br>[6.84]   |                   |                   |                   | 17.31<br>[6.14]   |
| Anomaly 4       |                   |                   |                   | -6.96<br>[-1.95]  |                   |                   | -0.37<br>[-0.08]  |
| Anomaly 5       |                   |                   |                   |                   | -8.56<br>[-3.03]  |                   | -7.73<br>[-2.30]  |
| Anomaly 6       |                   |                   |                   |                   |                   | -0.50<br>[-0.16]  | 13.82<br>[3.56]   |
| mkt             | -4.82<br>[-2.95]  | -4.72<br>[-2.80]  | -5.41<br>[-3.37]  | -5.51<br>[-3.33]  | -5.87<br>[-3.55]  | -5.55<br>[-3.33]  | -3.84<br>[-2.35]  |
| smb             | 24.52<br>[8.84]   | 16.18<br>[6.77]   | 8.30<br>[3.17]    | 18.60<br>[7.15]   | 18.13<br>[7.45]   | 16.82<br>[6.78]   | 14.97<br>[4.97]   |
| hml             | -15.96<br>[-3.12] | -33.03<br>[-9.08] | -27.94<br>[-8.17] | -29.89<br>[-5.84] | -31.44<br>[-8.39] | -36.94<br>[-8.53] | -11.49<br>[-2.06] |
| rmw             | 14.03<br>[4.34]   | 19.35<br>[5.84]   | 7.10<br>[2.05]    | 17.09<br>[5.30]   | 18.12<br>[5.61]   | 17.33<br>[5.29]   | 4.50<br>[1.19]    |
| cma             | 6.81<br>[1.47]    | 4.99<br>[1.07]    | 1.42<br>[0.31]    | 3.60<br>[0.76]    | 5.53<br>[1.18]    | 5.00<br>[1.06]    | 5.18<br>[1.12]    |
| umd             | 4.08<br>[2.51]    | 2.77<br>[1.69]    | 2.70<br>[1.69]    | 2.99<br>[1.82]    | 0.08<br>[0.04]    | 3.04<br>[1.75]    | 3.20<br>[1.63]    |
| # months        | 728               | 732               | 732               | 732               | 728               | 728               | 728               |
| $\bar{R}^2(\%)$ | 31                | 30                | 33                | 29                | 30                | 29                | 36                |



**Figure 8:** Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as SAC. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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